

Analyze Interoperability Standards and Smart Contract Platforms

This report examines blockchain interoperability standards and smart contract platforms as part of Weeks 3–4 of the thesis project plan. Drawing from 2025 literature and industry benchmarks, it analyzes mechanisms, implementations, and gaps relevant to the proposed cross-chain content rights management service on Polkadot. The focus aligns with the research question by evaluating standards and platforms that enable scalable, interoperable monetization for subscription, pay-per-view (PPV), and purchase models. Interoperability standards like IEEE 3221.01-2025 and protocols such as XCM, CCIP, and IBC facilitate cross-chain data/value exchange, while platforms like Ethereum (Solidity) and Polkadot (ink!) provide execution environments for rights-enforcing contracts.

In 2025, the interoperability market is projected to reach \$2.55 billion by 2029, increasing from \$0.7 billion in 2024. This growth is driven by enterprise requirements for seamless asset transfers. Smart contract platforms focus on Turing-complete execution, with Polkadot’s ecosystem supporting over 1,000 transactions per second via parachains. Addressing gaps in standardization and platform fragmentation, such as EVM silos, the project introduces XCM-integrated ink! contracts, facilitating efficient cross-chain rights verification.

1. Overview of Interoperability Standards

Interoperability standards establish protocols for cross-chain communication, ensuring atomicity, consistency, and security in transactions. They are classified into notary-based systems (centralized verification), relay-based systems (trusted hubs), and hash-locking mechanisms (HTLC for swaps). The primary standards for 2025 concentrate on enterprise adoption, with IEEE spearheading formalization efforts.

Table 1: Interoperability Standards — Description and Features

Standard / Protocol	Description	Key Features
IEEE 3221.01-2025	Cross-chain transaction consistency protocol for multiple technologies.	Notary/multi-sig verification; atomic swaps.
XCM (Polkadot)	Cross-consensus messaging for parachains.	Instruction-based (e.g., WithdrawAsset); multi-hop.
CCIP (Chainlink)	Oracle-enabled cross-chain protocol.	Risk management; embedded instructions.
IBC (Cosmos)	Inter-blockchain communication v2 (launched March 2025).	Packet-based messaging; sovereignty-preserving.
LayerZero	Omnichain protocol for arbitrary data.	Endpoint-based; gas abstraction.

Table 2: Interoperability Standards — Project Applicability

Standard / Protocol	Applicability to Project	Pros	Cons
IEEE 3221.01-2025	Rights transfers (e.g., PPV tokens across chains).	Standardized framework; reduces disputes.	Centralized elements in notary mode.
XCM (Polkadot)	Subscription activations via bridges.	Native to Polkadot; low latency (<2s).	Ecosystem-specific.
CCIP (Chainlink)	Secure PPV payments with oracles.	Modular security; 100+ chains.	Oracle dependency risks.
IBC (Cosmos)	Purchase verifications in multi-chain ecosystems.	115+ chains; trust-minimized.	Complex for non-Cosmos devs.
LayerZero	Hybrid subscription models.	Universal; developer-friendly.	Emerging; fewer audits.

2. Deep Dive into Interoperability Standards

IEEE 3221.01-2025: This IEEE standard delineates a framework for cross-chain consistency, encompassing notary-based mechanisms (e.g., multi-signature) and hash-locking techniques. It aims to resolve enterprise challenges such as transaction finality, with applications extending to supply chains and payment systems. In the year 2025, it is integrated with Polkadot to facilitate hybrid configurations, achieving 99% atomicity in asset exchanges.

XCM (Polkadot): As Polkadot’s native format, XCM v3 (updated 2025) utilizes instructions for asset transfers and remote execution, supporting NFTs and fungible tokens. It demonstrates excellence within shared-security ecosystems, with benchmarks in 2025 indicating a throughput exceeding 500 transactions per second across parachains.

CCIP (Chainlink): The 2025 version introduces AI-driven risk scoring for bridges, securing over \$10 billion in transfers. It simplifies complexity for developers, making it ideal for oracle-dependent rights verification.

IBC v2 (Cosmos): Launched at the end of March 2025, it enhances packet relaying capabilities for sovereign chains and has over 115 integrations. The focus is on minimal trust, facilitating cross-ecosystem royalties.

LayerZero v2: The 2025 updates incorporate omnichain fungibility, supported by over fifty blockchain networks. It highlights a gasless user experience suitable for micro-transactions such as Pay-Per-View (PPV).

Trends: In 2025, enterprises anticipate the implementation of hybrid standards, such as IEEE and XCM. However, challenges persist, including bridge hacks that resulted in a \$1.5 billion loss in 2024.

3. Overview of Smart Contract Platforms

Smart contract platforms offer runtimes for deterministic code execution. Ethereum maintains a dominant position with a 60% market share, while Polkadot’s ink! is gaining prominence for cross-chain applications. Comparative analyses in 2025 emphasize the trade-offs among speed, cost, and interoperability.

Table 3: Smart Contract Platforms — Technical Comparison

Platform	Language	TPS (2025 Avg.)	Cost/Tx	Interoperability
Ethereum	Solidity/Vyper	15–100 (L2: 2K)	\$0.01–0.10	Bridges (e.g., CCIP)
Polkadot (ink!)	Rust (ink!)	500–1K+ (parachains)	<\$0.01	Native XCM
Solana	Rust	2K–65K	<\$0.001	Wormhole bridges
BSC	Solidity	100–300	<\$0.01	BSC Bridge
Cardano	Haskell/Plutus	250–1K	\$0.10–0.50	Sidechains

Table 4: Smart Contract Platforms — Project Applicability

Platform	Applicability to Project	Pros	Cons
Ethereum	PPV contracts; EVM bridges.	Mature ecosystem; audited tools.	Congestion; high fees.
Polkadot (ink!)	Subscription/Purchase; cross-chain.	Secure (Rust); scalable.	Steeper learning curve.
Solana	High-volume PPV.	Ultra-fast; low cost.	Centralization risks.
BSC	EVM-compatible subs.	Cheap; Ethereum-like.	Centralization; security incidents.
Cardano	Formal verification for rights.	Secure; research-backed.	Slow development.

4. Deep Dive into Key Platforms

Ethereum (Solidity): Solidity v0.8.25 (2025) supports EVM Layer 2 solutions such as Optimism for enhanced scalability. It remains the predominant platform for decentralized finance and digital content, including OpenSea NFTs. However, the 2025 upgrades (Dencun) are projected to decrease Layer 2 transaction fees by approximately 90%. Interoperability is achieved through bridges; nonetheless, siloed ecosystems continue to exist.

Polkadot (ink!): ink! version 6 (2025) introduces Solidity compatibility via PolkaVM (RISC-V), facilitating migration to the EVM. Rust’s safety features prevent reentrancy, and parachains permit the development of custom VMs. Benchmark results indicate that it is ten times faster than Solidity when executing complex logic.

Solana: Developed in Rust, with the 2025 Firedancer upgrade anticipated to reach 1 million TPS. Suitable for media streaming contracts; however, outages such as the one projected for Q1 2025 underscore existing reliability concerns.

BSC & Cardano: BSC mirrors Ethereum for rapid implementation; Cardano’s Plutus emphasizes formal verification methods to ensure the security of intellectual property contracts.

Trends: 2025 sees multi-VM support (e.g., Polkadot’s EVM+ink!), with AI tools for auditing.

5. Gap Analysis

Analyses of 2025 indicate fragmentation: 40% of cross-chain transactions fail due to incompatible standards, and 30% of platforms lack native interoperability. These gaps impede project-like use cases, such as multi-chain subscriptions.

Table 5: Gap Analysis: Current Metrics vs Project Targets

Gap Area	Current Metric (2025 Avg.)	Project Target Improvement
Standard Adoption	25% enterprises (IEEE/IBC)	80% via XCM+IEEE hybrid
Cross-Chain Latency	5–10s (bridges)	<2s (native XCM)
Platform Security	15% vuln exploits (EVM)	99.9% (Rust/ink! audits)
Cost for Micro-Tx	\$0.05 (Ethereum)	<\$0.01 (Polkadot)
Dev Accessibility	60% Solidity-only	Multi-lang (ink!+Solidity compat)

Key Gaps: The absence of unified standards for content-specific interoperability, such as rights metadata, presents a significant challenge. Platform silos hinder scalability, exemplified by Ethereum’s base capacity of fewer than 100 transactions per second. The project addresses these issues through the development of XCM for standards-agnostic transfers and ink! for secure, cost-effective execution.

6. Relevance to the Thesis Project

- **Inform Design:** XCM/IBC map to cross-chain flows in SubscriptionManager.rs; ink! vs. Solidity justifies Polkadot choice for 60% cost savings.
- **Evaluation Metrics:** Benchmarks for interop (99% success) and scalability (500 TPS) in Phase 4.
- **Fills Gaps:** Native standards integration enables hybrid models, boosting economic efficiency vs. competitors.
- **Contributions:** Advances multi-VM platforms for content rights, addressing \$2.55B market needs.

Implementation Notes (Phase 4–5 Measured Results)

The theoretical benchmarks cited above were from 2025 literature. Phase 5 testing produced measured results that contextualise these predictions:

Table 6: Phase 1 Claims vs Phase 5 Measured Results

Phase 1 Claim	Measured Result	Notes
XCM latency “<2s”	18–32 seconds (3–5 parachain blocks)	HRMP relay through the relay chain adds latency not accounted for in the <2s theoretical estimate
Polkadot TPS “500–1K+ (parachains)”	27 TPS sustained per parachain; 283 TPS theoretical	Literature conflates aggregate network capacity (100 parachains $\times$ 27 $\approx$ 2,700 TPS) with single-parachain throughput
Cross-chain success “99%”	100% (9 XCM operations in testing)	Exceeded target, though with smaller sample size
ink! for “secure, cost-effective execution”	FRAME pallet as primary logic; ink! as API layer	Architecture pivoted for performance; ink! still used via pallet-revive precompile
Bridge latency “5–10s”	Snowbridge E2E: ~1 minute after beacon finalization	Full cryptographic proof verification adds latency vs simple bridge estimates

Bibliography

1. Liberty Street Economics. (2025). Interoperability of Blockchain Systems and the Future of Payments. *Federal Reserve Bank of New York*. <https://libertystreeteconomics.newyorkfed.org/2025/03/interoperability-of-blockchain-systems-and-the-future-of-payments/>

Annotation: In a previous publication, we introduced a tripartite framework for payment system interoperability that considers technological, legal, and economic factors to ensure the “singleness of money” and make payments consistent despite monetary fluctuations. Here, we extend this to the blockchain, propose an interoperability assessment method, demonstrate limited interoperability, and offer insights for future payment systems.

2. CryptoEQ. (2025). Blockchain interoperability: The current state in 2025. <https://www.cryptoeq.io/articles/blockchain-interoperability-solutions>

Annotation: Blockchain interoperability is crucial for a connected multi-chain ecosystem, allowing seamless transfer of assets, data, and functionalities. This report explores the necessity, challenges, and solutions, including token bridges, messaging networks, rollup mechanisms, Cosmos IBC, Polkadot XCMP, and emerging protocols such as Chainlink CCIP and LayerZero. It discusses security risks, governance, and trends such as standardization, zk-proof bridges, restaking, and improved user experience. As blockchain evolves, interoperability will drive adoption, liquidity, and a unified Web3 ecosystem. The report provides insights into the future of cross-chain infrastructure for developers, investors, and the crypto industry.

3. IBC Protocol. (2025). IBC v2: Enabling IBC Everywhere. <https://ibcprotocol.dev/blog/ibc-v2-announcement>

Annotation: IBC v2 builds on IBC, simplifying the protocol while preserving its expressiveness. It eases connection with various blockchains, starting with Ethereum and Solana. A streamlined design reduces complex handshakes, lowering implementation efforts and enabling faster cross-chain links, even for ecosystems without light clients. ZK verification cuts costs, making Ethereum connections affordable and secure, with IBC Ethereum links launching by March 2025.

4. Phoenix Strategy Group. (2025). Ultimate guide to blockchain interoperability for enterprises. <https://www.phoenixstrategy.group/blog/ultimate-guide-to-blockchain-interoperability-for-enterprises>

Annotation: Blockchain interoperability enables networks to share data, transfer value, and communicate without intermediaries, which is vital for multi-platform businesses. Challenges include legacy systems, scalability, cross-chain issues, and U.S. regulatory uncertainty with evolving SEC and state rules. Solutions involve skilled teams creating APIs, middleware, ensuring AML/KYC compliance, and system upgrades, supported by advisory services. Key protocols include Polkadot with a secure Relay Chain, Cosmos with decentralization, and others such as Basel Protocol and Hyperlane for legacy systems. Future trends focus on asset tokenization, ESG, AI, and CBDCs.

5. Mor Software. (2025). Blockchain interoperability: Ultimate guide for enterprises 2025. <https://morsoftware.com/blog/blockchain-interoperability>

Annotation: As blockchain networks grow, the need for seamless blockchain interoperability becomes urgent. Many enterprises face isolated systems that restrict data exchange, raise costs, and hinder innovation. This MOR Software guide explores how interoperability blockchain solutions are transforming enterprises in 2025, boosting connectivity and growth.

6. ScienceDirect. (2025). Towards blockchain interoperability: a comprehensive survey on cross-chain solutions. <https://www.sciencedirect.com/science/article/pii/S2096720925000132>

Annotation: The rapid growth of DeFi apps has spurred the development of new blockchain systems that mostly evolve in isolation, limiting seamless value and data flows. Interoperability technologies aim to remove these barriers for effective cross-network interactions. While some solutions have been proposed and classified by standards, deeper analysis is needed. This work reviews mainstream cross-chain solutions, covering their principles, applications, protocols, and performance. A conceptual model clarifies asset and data interoperability, with a hierarchical architecture categorizing key solutions from academia and industry. It also highlights open challenges and future directions, providing a comprehensive overview of cross-chain solutions.

7. CoinLaw. (2025). Blockchain Interoperability Statistics 2025: Market Growth, Challenges, and Future Outlook. <https://coinlaw.io/blockchain-interoperability-statistics/>

Annotation: Imagine a world where blockchains collaborate, each leveraging its strengths. By 2025, interoperability will become real, transforming finance to healthcare. Growing

adoption drives the need for secure, efficient cross-chain data and asset transfers. This article explores the current state of interoperability and key stats.

8. Metana. (2025). Top 9 smart contract programming languages in 2025. <https://metana.io/blog/top-9-smart-contract-programming-languages/>

Annotation: Solidity/Rust comparison. Guides ink! choice; Ethereum bias.

9. Rapid Innovation. (2025). Top 12 blockchain platforms for smart contract development in 2025. <https://www.rapidinnovation.io/post/top-12-blockchain-platforms-for-smart-contract-development>

Annotation: Blockchain is transforming industries such as finance, healthcare, and supply chains, with the global market expected to reach \$94 billion by 2027. Central to this are smart contracts—self-executing agreements that enforce terms without intermediaries—powering DeFi, NFTs, and more. This blog highlights the top 12 blockchain platforms redefining smart contract development in 2025, with a focus on transaction speed and consensus mechanisms. Whether you're a developer, investor, or enthusiast, this guide provides key insights into the best platforms for smart contracts.

10. Our Crypto Talk. (2025). Top 10 smart contract platforms in 2025. <https://web.ourcryptotalk.com/blog/top-10-smart-contract-platforms-in-2025>

Annotation: Smart contract platforms are transforming the way digital agreements are executed. These self-executing contracts, stored on a blockchain, automatically perform actions when predefined conditions are met no banks, brokers, or middlemen required. They've become the backbone of decentralized finance (DeFi), NFTs, gaming, supply chain automation, and even corporate governance.

This guide explores the Top 10 Smart Contract Platforms in 2025, breaking down their scalability, security, developer support, and ecosystem maturity to help you choose the one best suited for your goals.

11. NDLabs. (2025). Top smart contract platforms in 2025: A comprehensive guide. <https://ndlabs.dev/top-smart-contract-platforms>

Annotation: Polkadot ecosystem details. Informs parachain use; interop emphasis.

12. Calibraint. (2025). The 7 Most Powerful Smart Contract Platforms to Build and Scale in 2025. <https://www.calibraint.com/blog/top-smart-contract-platforms-list-2025>

Annotation: In 2025, Smart Contracts are essential for businesses to streamline operations, cut costs, and improve security. They are transforming industries like supply chain, finance, real estate, gaming, and healthcare. The Smart Contract market was valued at USD 2.2 billion in 2024 and is projected to reach USD 29.2 billion by 2035, with a CAGR of 26.5%, underscoring the need for businesses to adopt Smart Contracts. Several Smart Contract Platforms exist, each prioritizing different features such as speed, cost, security, or privacy. Choosing the right platform depends on your goals, industry, and scalability. How to select the best fit? We will discuss this in this guide.

13. Minddeft. (2025). Most popular blockchain development tools in 2025. <https://minddeft.com/blog/most-popular-blockchain-development-tools>

Annotation: Blockchain technology is advancing rapidly, with Solidity, Vyper, and Rust as top smart contract choices in 2025. Ethereum, Solana, and Polkadot shape the ecosystem, while Truffle, Hardhat, and Ganache aid testing. Etherscan and The Graph provide analytics. These tools support dApps, DeFi, and enterprise solutions, keeping developers ahead.

14. TechTarget. (2025). Top 9 smart contract platforms to consider in 2025. <https://www.techtarget.com/searchcio/tip/Top-smart-contract-platforms-to-consider>

Annotation: Smart contracts allow users to write executable code that encodes business logic on a blockchain or decentralized ledger. Many deployment options exist, supporting the secure, transparent, tamper-proof execution of predefined conditions for trustless applications such as finance, supply chain, and voting. There are four main smart contract platform types: large blockchain platforms, smaller ones optimized for speed or cost, private frameworks, and specialized tools for contract creation with trusted partners. These were chosen based on prominence, expert insights, market cap, and trading volume.

15. CryptoSlate. (2024/2025). Comparing smart contracts across different blockchains from Ethereum to Solana. <https://cryptoslate.com/are-all-smart-contracts-created-equal-how-top-turing-complete-blockchains-compare/>

Annotation: The crypto industry has unique approaches to smart contracts and DApps, driven by needs for scalability, security, and efficiency, enabling sophisticated applications. Turing completeness indicates a system's ability to perform any computation given enough resources. Leading platforms like Ethereum, ICP, Polkadot, Cardano, and Solana each use Turing completeness in different ways. This article examines how they address challenges in the decentralized ecosystem.

16. Crypto Adventure. (2025). Ethereum vs. Other Smart Contract Platforms: A Comprehensive Comparison. <https://cryptoadventure.com/community/articles/ethereum-vs-other-smart-contract-platforms-a-comprehensive-comparison/>

Annotation: Ethereum pioneered the concept of programmable smart contracts and remains the most widely used platform in the blockchain ecosystem. However, over the years, several alternative smart contract platforms have emerged to address Ethereum's limitations, offering faster performance, lower fees, enhanced scalability, and different consensus models. This article provides an in-depth comparison between Ethereum and its major competitors in 2025, including Solana, Binance Smart Chain (BSC), Avalanche, Cardano, and Polkadot.

17. Cubix. (2025). Top 10 smart contract platforms dominating 2025. <https://www.cubix.co/blog/top-smart-contract-platforms/>

Annotation: This post will explore the top 10 smart contract platforms expected to dominate in 2025. We'll examine what makes each platform unique, highlighting its strengths and contributions to blockchain development. As a blockchain development company, Cubix works closely with these platforms to create advanced dApps and smart contract solutions.

18. Gates AI. (2025). IEEE Std 3221.01-2025: IEEE Standard for Blockchain Interoperability — Cross-Chain Transaction Consistency Protocol. <https://www.gates-ai.com/ieee-std-3221-01-2025-ieee-standard-for-blockchain-interoperability-cross-chain-transaction-consistency-protocol/>

Annotation: As blockchain technology advances, many networks emerge, but their lack of direct value and of mechanisms for information exchange creates silos, hindering adoption. Without effective interoperability, frequent reconfigurations are needed. To solve this, the IEEE Computer Society’s Blockchain and Distributed Ledger Standards Committee released IEEE Standard 3221.01-2025, a high-performance protocol for cross-chain transaction consistency. This standard promotes interoperability, enhances security, and fosters a collaborative blockchain ecosystem, unlocking the full potential of blockchain.